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Effects of ternary alloy additions on the microstructure of highly immiscible Cu alloys subjected to severe plastic deformation: An evaluation of the effective temperature model

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ABSTRACT

Phase evolution in dilute, strongly immiscible Cu-Mo, Cu-Mo-Ni, and Cu-Mo-Ag alloys during severe plastic deformation at low-temperature has been experimentally investigated. For the Cu₉₅Mo₀₅ alloy, Mo nanoparticles are formed, ~10 nm in diameter, as part of a steady state microstructure, with less than ~1 at.% Mo dissolved in the matrix. Addition of 10 or 20 at.% Ni to this binary alloy results in a significant increase in the Mo solubility, whereas comparable additions of Ag has a corresponding little effect. The steady state microstructures of alloys during ball milling of elemental powders are very similar to those during HPT processing of initially homogeneous solutions. The results are discussed in terms of an effective temperature model. Model MD simulations are presented to help relate the predictions of the effective temperature model to atomistic mechanisms.

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1. Introduction

Severe plastic deformation (SPD) of metal alloys often leads to the chemical mixing of alloying constituents, commonly referred to as mechanical alloying [1]. This is true for most immiscible alloys as well, as long as the processing temperature is low enough to suppress thermally activated diffusion [2,3]. This mixing behavior in immiscible alloys has been rationalized in terms of a competition between forced chemical mixing tending to homogenize an alloy, and thermally activated diffusion tending to restore equilibrium [4]. Similar behavior has been observed during irradiation of immiscible alloys, and indeed, it has been explained on the same basis, the main difference involving the details of the forced mixing process. Martin [5] and Martin and Bellon [6] generalized this dynamical behavior by suggesting that the forcing term, which randomizes atomic positions, could be considered as an infinite

temperature process, or equivalently as an introduction of configurational entropy into the alloy. The system under atomic forcing in this picture can be described in terms of an effective free energy, which corresponds to true free energy, but at a corresponding higher effective temperature,

$$T_{\text{eff}} = T(1 + \gamma_b^\infty), \quad (1)$$

than the actual processing temperature, T [5,6]. Here $\gamma_b^\infty = D_b^\infty/D_{\text{th}}$ represents the ratio of diffusion coefficients arising from ballistic (i.e., random) atomic jumps and thermally activated atomic jumps (usually mediated by point defects). At low temperature $D_{\text{th}} \rightarrow 0$ and $T_{\text{eff}} \rightarrow \infty$, and for this reason, alloys tend to homogenize.

Several works have shown, however, that *strongly* immiscible alloys do not homogenize during forced chemical mixing at room temperature, or even at much lower temperatures, either by irradiation [7], or by SPD [8–11]. In the case of irradiation, the mechanism controlling phase separation has been attributed to thermal spikes created by energetic displacement cascades, i.e., as an ion slows in a metallic target, it dissipates several hundreds of eV's to keV's in a localized region in less than a picosecond. As the system

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locally equilibrates over the next several ps, this region undergoes local melting [12]. Since the liquid-like region persists for ~ 10 ps, and the diffusion coefficient of liquid metals is $\sim 1 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$, every atom can diffuse one or two atomic distances within the thermal spike. Thus, in alloys that have a very limited solubility in the melt, precipitation can take place after several such overlapping events. Additional details are found in Refs. [13–15], but notice that the ballistic mixing in this situation is not random, as in the Martin model, rather, it is characteristic of the thermal spike temperature. The atomic relocation processes during shear-induced mixing are not nearly as well understood at present, particularly in strongly immiscible alloys where precipitates are often incoherent and the elastic properties of the precipitate and matrix can be very different. Clearly, however, thermal spikes like those encountered during ion irradiation do not explain the lack of homogenization during SPD. On the other hand, SPD involves local stresses that force atoms over potential barriers and strong interatomic interactions are likely to bias this motion. Such biased relocation of atoms during low-temperature SPD is suggested by several experiments [16,17]. In Ref. [17], for example, *homogeneous, solid-solutions* of dilute Cu-Nb alloys subjected to high pressure torsion (HPT) at -78°C were observed to undergo precipitation of Nb and nearly complete loss of Nb from solution. The very low-temperature processing employed in these experiments, moreover, precludes the possibility of shear-enhanced thermally-activated diffusion since vacancies are virtually immobile in Cu at this temperature.

Lund and Schuh attempted to account for the biased motion of atoms during shear within the Martin effective temperature model by including it as a contribution to the *thermally activated diffusion* [18]. In this modified model, the effective temperature is,

$$T_{\text{eff}} = T \left(1 + D_b^\infty / \left(D_{\text{th}} + D_b^\infty \frac{T}{T_0} \right) \right) \quad (2a)$$

or in terms of γ_b^∞

$$T_{\text{eff}} = T \left(1 + \gamma_b^\infty / \left(1 + \frac{T}{T_0} \gamma_b^\infty \right) \right), \quad (2b)$$

where now $T_{\text{eff}} \rightarrow T_0$ as $T \rightarrow 0 \text{ K}$, T_0 being a temperature characterizing the ballistic dynamics in forcing the mixing of elements. Lund and Schuh further employed computer simulation to illustrate the correspondence between a system driven at 0 K (using molecular statics) with a system in equilibrium at temperature, T_{eff} , (using Monte Carlo), by comparing the chemical short-range order parameter for the two systems as a function of the strength of the interatomic interaction, i.e., the heat of mixing in a binary alloy. The systems considered by Lund and Schuh were two-dimensional. Later Ashkenazy et al. showed a similar correspondence in alloy microstructure during low temperature SPD and high temperature equilibration using large-scale 3-d molecular dynamics (MD) simulations of *amorphous* Cu-Ta [19]. While Lund and Schuh do not provide a rigorous justification for the form of eq. (2), we show in the Appendix that by replacing the infinite-temperature ballistic diffusion coefficient, D_b^∞ in the original Martin model (eq. (1)), with a ballistic diffusion coefficient characteristic of the effective process temperature, $D_b^{T_0}$, (in the spirit of Lund and Schuh) a new effective temperature can indeed be derived,

$$T_{\text{eff}} = T \left(1 + \gamma_b^{T_0} / \left(1 + \frac{T}{T_0} \gamma_b^{T_0} \right) \right) \quad (3)$$

where now we define the forcing intensity as $\gamma_b^{T_0} = D_b^{T_0} / D_{\text{th}}$. Note, $D_b^{T_0}$ does not result in random displacements as does D_b^∞ in eqs. (1) and (2), just as diffusion in thermal spikes during irradiation need

not be random. At low processing temperatures, $T < T_0$, $T_{\text{eff}} \rightarrow T_0$, and at high processing temperatures, $T > T_0$, $T_{\text{eff}} \rightarrow T$, and the new effective temperature agrees with that of Lund and Schuh. Between these two limits, however, the two models behave somewhat differently, as shown in Fig. 1.

Since the experiments presented here are conducted at a single temperature, close to the low-temperature limit, we will not attempt to distinguish these two models, further. The present work considers, rather, whether such modified effective temperature models are, in any case, useful in understanding forced chemical mixing in systems involving strong chemical interactions. We do this by examining the effect of the heat of mixing on the steady state microstructure of alloys subjected to SPD. While the heat of mixing of an alloy cannot be varied systematically while keeping other alloy properties fixed, as done by Lund and Schuh using computer simulation, we approximate this situation by measuring the steady state solubility of Mo in dilute $(\text{Cu}_{1-x}\text{Ni}_x)_{1-y}\text{Mo}_y$ and $(\text{Cu}_{1-x}\text{Ag}_x)_{1-y}\text{Mo}_y$ alloys during SPD using ball milling and high-pressure torsion (HPT). In these experiments, $0 \leq x \leq 0.36$ and $0.01 \leq y \leq 0.06$. These alloys are selected since Mo is highly immiscible in Cu, the heat of solution being $>100 \text{ kJ/mol}$ (in dilute Cu-Mo alloys), while Ni is highly soluble in Cu, and Mo is highly soluble in Ni. Adding Ni to the Cu-Mo alloy, therefore, lowers the effective heat of mixing of Mo in the Cu rich matrix. Ag, like Ni, has high solubility in Cu, but it is highly immiscible with Mo, and thus its addition to the alloy does not change the effective heat of mixing considerably. We discuss the thermodynamics properties of these alloys in more detail below. Inherent to effective temperature models is a unique steady-state alloy configuration. As part of this work, therefore, we also examine the steady state configuration starting from two very different initial states, (i) a mixture of Cu and Ni-Mo alloy powders using ball milling and (ii) homogenous solutions of Cu-Ni-Mo and Cu-Ag-Mo using HPT.

2. Methods

2.1. Experimental details

Ball milling experiments were performed under a controlled Ar atmosphere using a Spex 8000 mixer mill, either nominally at room temperature (RT) (actual temperature $T \sim 40^\circ \text{C}$) or at $T \sim -15^\circ \text{C}$ (cryo-milling). Cryo-milling directs cold nitrogen gas toward the

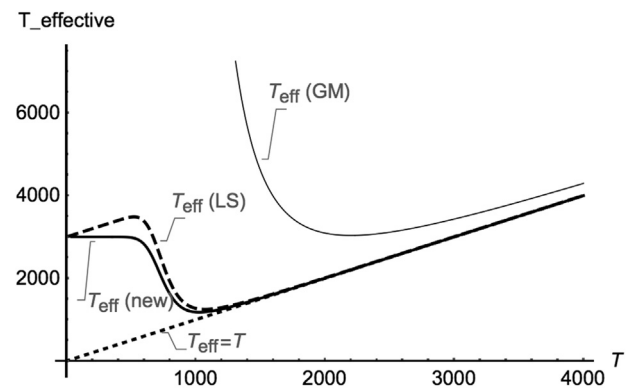


Fig. 1. Plot of various effective temperatures as a function of the actual temperature T using Martin's model [5], labeled as GM, Lund and Schuh's heuristic expression [18] labeled as LS, and the new expression eq. (3) (derived in the appendix), labeled as "new". The asymptotic behavior expected at high temperature, $T_{\text{eff}} = T$, is represented as a dotted line. In this plot, the vacancy migration energy was set to 0.8 eV , the effective process temperature to 3000 K , the forcing intensities γ_b to 10^{-3} and $\gamma_b^{T_0}$ to 10^{-5} .

mixing vial during the milling process. The Cr-steel milling balls to powder ratio (by mass) was 5g:25g. Ag powder was added to all Cu mixtures to prevent cold welding [16]. Since Ag is highly immiscible with Mo, like Cu, it has little effect on the alloying process (as discussed in more detail below). A baseline for Mo solubility in Cu was obtained by milling $\text{Cu}_{89}\text{Mo}_{05}\text{Ag}_{06}$ alloys both at RT and during cryo-milling. Ball milling experiments of ternary Cu-Mo-Ni alloys (plus ~6 at.% Ag) were performed in a two-step process: Ni and Mo were first mechanically alloyed at RT for 10 h and then the product was milled with Cu(Ag). Several mixtures of Ni and Mo powders were milled to create Ni-Mo alloys with varying composition, see Table 2. The Ni and Mo powder were first pre-mixed in these experiments, rather than simply milling the three constituent metal powders, to gain better control over the Ni:Mo ratio, as it is difficult to add accurately just a few percent Mo to the Cu alloy. Previous work on Cu-Ag-Ni showed that the order of mixing the powders had no effect on the alloy microstructure in steady state [20]. We will also show this insensitivity to the initial state of the alloy by comparing results using ball milling and high pressure torsion.

Powders were characterized by X-Ray diffraction (Siemens-Bruker D-5000) using Cu K_α radiation for phase identification and measuring solubilities. Precise lattice parameters were obtained with the help of Rietveld analysis using the GSAS package [21]. For calibration purposes, small quantities of pure Cu and Ni powders were added to the milled specimens. Calculated lattice parameters were then used to estimate the solubility of Mo in the Cu phase using empirical modifications to Vegard's law. Fe contamination from the milling medium (typically ~1 at.%) was quantified using X-Ray fluorescence (XRF, Shimadzu EDX-7000), and its contribution to the lattice parameter was factored out. Transmission electron microscopy (TEM) (JEOL 2010F (S)TEM; JEOL 2010 LaB₆ TEM; JEOL 2200 FS (S)TEM; FEI Themis Z) was used to establish the alloy/secondary phase grain size/morphology and residual solubility. TEM samples were fabricated by focused ion beam erosion (FIB, FEI Helios 600i), using Ga ions. Dark field imaging was employed to determine precipitate sizes for all of the alloys.

In addition to the ball milling experiments, a separate set of samples was synthesized for HPT experiments using magnetron sputtering. These samples consisted of dilute solid solutions of Cu-Mo, Cu-Ni-Mo and Cu-Ag-Mo. The sputtering was carried out using AJA ST-20 DC magnetron sputtering sources. The base pressure before sputtering was $<1 \times 10^{-8}$ Torr, the sputtering target purity was 4 N, and the Argon sputtering gas was 99.999% pure. Since HPT requires specimens greater than ~100 μm thick, the sputtered films were grown to a thickness of ~20 μm on 3" oxidized Si wafers. The films were subsequently stripped from the wafers, broken into small flakes, and consolidated in the HPT at RT. A thin Ag layer was deposited on the Si wafer prior to depositing the alloy and another thin Ag layer (~20–40 nm) was deposited on the surface of the specimen. These layers prevented oxidation of the specimen and assisted in the consolidation process. The HPT processing was performed at dry ice temperature (~78.5 °C) in an unconstrained configuration, i.e., the compacted disk was placed between flat

anvils and subjected to uniaxial torsion under a pressure of 4.5 GPa for up to 40 cycles. A cycle represents a rotation of the anvils 90° relative to each other in one direction and then back; the rotation speed was 1 rpm. The strain increases from zero at the center of the specimen to a maximum at $R = 5\text{--}6\text{ mm}$ (radius of the disk) with the local strain given by $\gamma = \pi nr/h$ where n is the number of cycles, r is the radial position, and h is the local thickness of the disk at location r . To determine the strain accurately, the disk was cut in half and the sample thickness was measured as a function of r . The analysis was restricted to radial positions with $r < 3.5\text{ mm}$ to avoid possible edge effects. The HPT processed samples were analyzed by X-ray diffraction (PANalytical/Philips X'Pert) in a micro diffraction mode, i.e., a micro-capillary system is used to focus the beam to a diameter of ~100 μm ; a camera is used to accurately select the region of interest for analysis. Prior to X-ray analysis, the samples were polished to remove possible surface damage. Samples employed for TEM analysis were selected from the middle of the cross section of the disks, also to avoid possible surface artifacts.

2.2. Molecular dynamics and Monte Carlo simulations

We performed a combination of molecular dynamics (MD) and Monte Carlo (MC) simulations to examine the modified effective temperature model in 3-d Cu alloys, similar to the method employed by Lund and Schuh on 2-d Lennard-Jones (L-J) systems [18] and Ashkenazy et al. on amorphous Cu-Ta [19]. Basically, our method determines the equilibrium short range order parameter, Ω_{eq} , as a function of temperature using MC, and then locates the temperature, T_{eff} , by finding where $\Omega_{\text{eq}}(T) = \Omega_{\text{SS}}^{\text{SPD}}$; $\Omega_{\text{SS}}^{\text{SPD}}$ is the short range order parameter during low-temperature ($T = 100\text{ K}$) shearing in steady state. We use here the short range order parameter defined by Lund and Schuh [18],

$$\Omega = \frac{\frac{c}{1-c}N_A^B + \frac{1-c}{c}N_B^A}{N_A^A + N_B^B} \quad (4)$$

where N_A^B is the average number of B atoms around A atoms and c is the atomic fraction of A atoms. Ω has the value $-1, 0, 1$ for microstates that are fully decomposed, random, or fully ordered.

The code LAAMPS [22] is employed for both the MC and MD simulations. The MD system contains 32,000 atoms, and it is deformed using cyclic biaxial compression at 100 K (using a Nose-Hoover thermostat) and a strain rate of $5 \times 10^9\text{ s}^{-1}$. The exact strain path and other details are found in the Supplementary Information. The MC/MD code in LAAMPS was also employed to determine the equilibrium solubility Mo in Cu and the Cu-Ni alloys as a function of temperature; these details too are reported in the Supplementary Information.

Cleri-Rosato potentials are used in the present simulations to represent Cu [23]. Like L-J potentials, the form of these potentials allows the heat of mixing of an alloy to be varied systematically, while keeping the elastic properties nearly unchanged. Specifically, we create $(\text{Cu}(1)_{1-x}\text{Cu}(2)_x)_{1-y}\text{Cu}(3)_y$ alloys where the interatomic

Table 1
Mo solubility for all alloys processed by ball milling and HPT.

Ball Milled Powders						HPT Samples			
Sample	Mo Sol (at.%)	Sample (RT)	Mo Sol (at.%)	Sample (RT)	Mo Sol (at.%)	Sample (RT)	Mo Sol (at.%)	Sample (−78.5 °C)	Mo Sol (at.%)
$\text{Cu}_{89}\text{Mo}_{05}\text{Ag}_{06}$ (RT)	0.3	$\text{Ni}_{90}\text{Mo}_{10}$	10	$\text{Cu}_{84}\text{Ni}_{09}\text{Mo}_{01}\text{Ag}_{06}$	1	$\text{Cu}_{74}\text{Ni}_{18}\text{Mo}_{02}\text{Ag}_{06}$	2	$\text{Cu}_{95}\text{Mo}_{05}$	0.8
$\text{Cu}_{89}\text{Mo}_{05}\text{Ag}_{06}$ (CryoT)	0.6	$\text{Ni}_{80}\text{Mo}_{20}$	17	$\text{Cu}_{84}\text{Ni}_{08}\text{Mo}_{02}\text{Ag}_{06}$	1.7	$\text{Cu}_{64}\text{Ni}_{27}\text{Mo}_{03}\text{Ag}_{06}$	3	$\text{Cu}_{85}\text{Mo}_{05}\text{Ni}_{10}$	2.3
		$\text{Ni}_{70}\text{Mo}_{30}$	25	$\text{Cu}_{84}\text{Ni}_{07}\text{Mo}_{03}\text{Ag}_{06}$	2.3	$\text{Cu}_{54}\text{Ni}_{36}\text{Mo}_{04}\text{Ag}_{06}$	4	$\text{Cu}_{75}\text{Mo}_{05}\text{Ni}_{20}$	2.7
						$\text{Cu}_{72}\text{Ni}_{18}\text{Mo}_{04}\text{Ag}_{06}$	2.4	$\text{Cu}_{75}\text{Mo}_{05}\text{Ag}_{18}$	1
						$\text{Cu}_{70}\text{Ni}_{18}\text{Mo}_{06}\text{Ag}_{06}$	2.6		

Table 2

Mo particle size and density for all the binary and ternary alloys.

Ball Milled Powders						HPT Samples			
Sample	Particle Size (nm) [Density ($10^3/\mu\text{m}^3$)]	Sample (RT)	Particle Size (nm) [Density ($10^3/\mu\text{m}^3$)]	Sample (RT)	Particle Size (nm) [Density ($10^3/\mu\text{m}^3$)]	Sample (RT)	Particle Size (nm) [Density ($10^3/\mu\text{m}^3$)]	Sample (−78.5 °C)	Particle Size (nm) [Density ($10^3/\mu\text{m}^3$)]
Cu ₈₉ Mo ₀₅ Ag ₀₆ (RT)	12 [30]	Ni ₉₀ Mo ₁₀ 0		Cu ₈₄ Ni ₀₉ Mo ₀₁ Ag ₀₆ 0		Cu ₇₄ Ni ₁₈ Mo ₀₂ Ag ₀₆ 0		Cu ₉₅ Mo ₀₅	11 [36]
Cu ₈₉ Mo ₀₅ Ag ₀₆ (CryoT)	13 [22]	Ni ₈₀ Mo ₂₀ 11 [25]		Cu ₈₄ Ni ₀₈ Mo ₀₂ Ag ₀₆ 8 [7]		Cu ₆₄ Ni ₂₇ Mo ₀₃ Ag ₀₆ 0		Cu ₈₅ Ni ₁₀ Mo ₀₅	10 [30]
		Ni ₇₀ Mo ₃₀ 13 [24]		Cu ₈₄ Ni ₀₇ Mo ₀₃ Ag ₀₆ 10 [9]		Cu ₅₄ Ni ₃₆ Mo ₀₄ Ag ₀₆ 0		Cu ₇₅ Ni ₂₀ Mo ₀₅	9 [28]
						Cu ₇₂ Ni ₁₈ Mo ₀₄ Ag ₀₆ 11 [13]			
						Cu ₇₀ Ni ₁₈ Mo ₀₆ Ag ₀₆ 12 [21]			

potential describing Cu(i) is that of pure Cu, but the cross terms in the potential describing the interactions between the different Cu(i) are altered to yield different heats of mixing. This procedure has been used previously in binary alloys for a similar purpose [24]. While this procedure does not capture the variations in the properties of the pure materials, such as the difference in hardness and structure between Cu and Mo, it isolates the role of the heat of mixing on alloy microstructure during SPD at low temperatures. Details of the Cleri-Rosato potentials used here are described in Ref. [24].

3. Results

3.1. Ball milling results

3.1.1. Cu(Ag)-Mo alloy

X-ray diffraction results shown in Fig. 2 reveal that after RT- or cryoT-ball milling for 10 h the Cu-Mo binary system remains largely phase separated. We estimate the solubility limit of Mo in Cu during ball milling by evaluating the change in the lattice parameter. A large shift in the lattice parameter from pure Cu is indeed observed in Fig. 2, but this shift is due predominantly to the mixing of Ag into the Cu as Ag is completely miscible with Cu during RT- and CryoT-ball milling [3,25]. The contribution of Ag to the change in lattice parameter was obtained using XRF to determine the precise Ag concentration, coupled with the lattice parameter data for Cu-Ag alloys from Ref. [26]. With this correction, and calibrating the change in lattice parameter with additions of Mo, see below, we

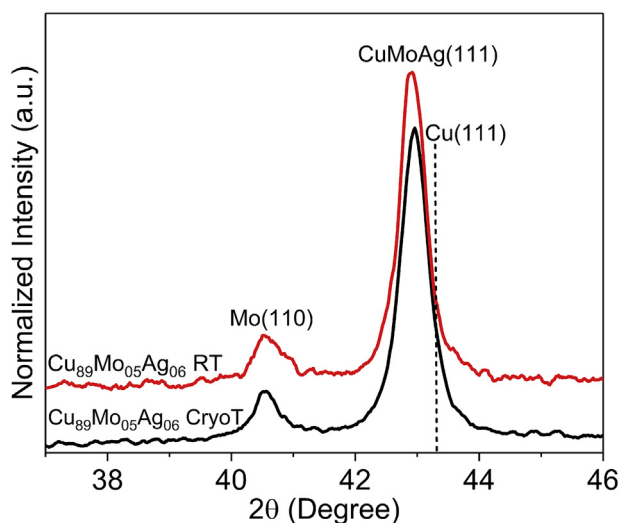


Fig. 2. X-Ray diffraction patterns for Cu-Mo ball-milled at room temperature (RT) and cryo-temperature (CryoT). The shift in the CuMoAg solution (111) peak, relative to the pure Cu peak, is mainly due to the dissolution of Ag, as the Mo solubility is small.

find that the solubility of Mo in Cu is ~0.3 at.% and ~0.6 at.% for RT and cryo-milling, respectively (Table 1). These values lie just outside the uncertainty of the measurements ± 0.3 at.%. The position of the Mo peak, moreover, shows that the Mo phase remains BCC after milling, with a negligible solubility of Cu in Mo.

Typical STEM HAADF images of the ball-milled Cu₈₉Mo₀₅Ag₀₆ samples are shown in Fig. 3(a) and (b) for the RT and CryoT-milled specimens, respectively. The bright features in these figures are identified by Energy Dispersive X-Ray Spectroscopy (EDX) analysis and Dark Field imaging to be Mo; these Mo particles show a uniform distribution throughout the Cu-Ag matrix. No Ag particles are found, as expected [25]. EDX also reveals the presence of Ag and traces of Mo in the Cu grains; the Mo signal, however, is too weak to quantify its concentration, other than to set an upper limit on its solubility at ~1 at.%. The size distribution of the Mo particles is reported in Fig. 4; the average particle diameter is 12 ± 5 nm for RT and 13 ± 4 nm for cryo-milling. The Mo particles are thus reduced in size by shearing, from ~30 μm in the initial powder to ~10 nm after milling. Strong refinement of highly immiscible precipitates during RT- and low temperature SPD is commonly observed [see e.g. Ref. [11]]. No indication of Mo-particle fracture is observed; rather the smooth, equiaxed shapes seen in Fig. 3 suggest that an erosion process at the Cu/Mo interface controls the morphology of Mo precipitates as they approach their steady-state size. For details of the erosion process see e.g., Refs. [27,28].

3.1.2. Mo-Ni binary alloy at RT

As noted above, Mo and Ni were co-milled before ball milling with Cu. X-ray diffraction of the Ni:Mo (9:1) alloy, Fig. 5, indicates that ball milling of Ni and Mo leads to a homogeneous alloy, as expected from previous work [29]. As the molar fraction of Mo was increased to 0.20 and 0.30, however, a BCC Mo diffraction peak begins to emerge and grow, also shown in Fig. 5. The solubility of Mo in Ni was calculated using Vegard's law, yielding solubilities of 17 at.% and 25 at.% for Ni₈₀Mo₂₀ and Ni₇₀Mo₃₀, respectively (Table 1). Additional evidence for the lack of complete solubility is obtained by DF imaging of the Ni-Mo alloys, which reveals Mo precipitates (Fig. S1 in Supplementary Information). The average size of these Mo particles in the Ni₈₀Mo₂₀ and Ni₇₀Mo₃₀ alloys is $\sim 13 \pm 3$ nm (see Fig. 6(a)), thus comparable to the size observed for Mo particles in the Cu-Mo alloy (Fig. 4). Also similar to ball milling of Cu-Mo is the absence of a measurable shift in the Mo(110) peak position, implying little mixing of Ni into the Mo phase (within the uncertainty ± 0.6 at.%). For reference, the maximum equilibrium solubility of Mo in Ni (FCC) is ~13 at.% at RT, and 27 at.% at the eutectic temperature (1317 °C), whereas negligible solubility of Ni is found on the Mo-rich side of the phase diagram at RT, and about 1 at.% at the melting temperature of Ni [29]; we return to this point below. These general observations agree well with ref. [29,30], where a maximum of 23–27 at. % of Mo dissolved in Ni was reported for Ni₆₀Mo₃₀ compositions milled for 40 h.

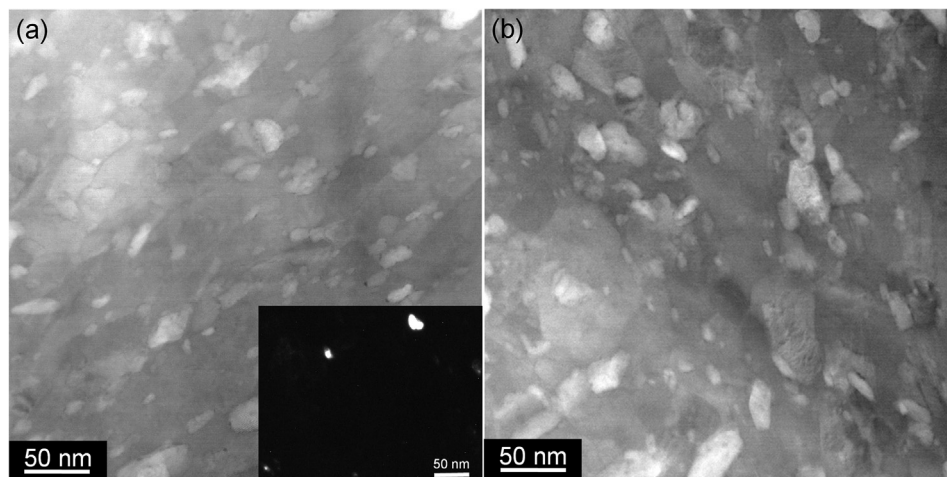


Fig. 3. High angle annular dark field (HAADF) taken in Scanning transmission electron microscopy (STEM) mode of RT (a) and Cryo temperature (b) specimens, where the bright contrast is mostly due to the higher Z of Mo. Dark field images taken by selecting a diffraction spot of Mo, as shown in the inset, confirm the Mo size and distribution.

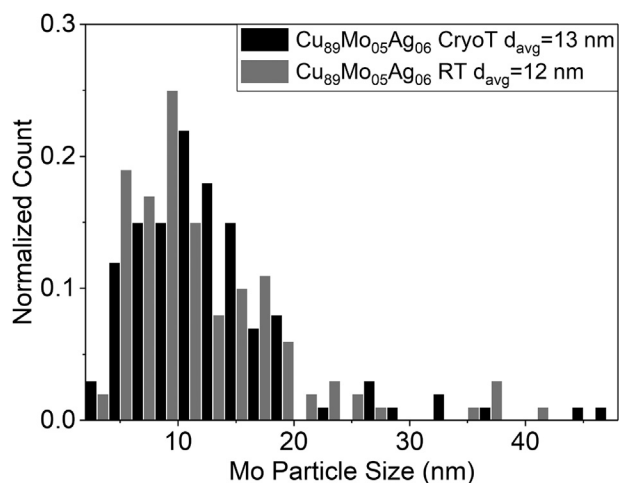


Fig. 4. Particle size distribution of Mo in $\text{Cu}_{89}\text{Mo}_{05}\text{Ag}_{06}$ for RT and CryoT.

3.1.3. Cu(Ag)-Mo-Ni ternary alloy at RT

The ternary alloys (plus Ag) were prepared by milling together the pre-alloyed Ni-Mo with Cu-6 at.% Ag. Several compositions were examined, see Table 1, to provide a systematic understanding of the roles of both Ni and Mo concentrations in determining the solubility of Mo in Cu. Increasing the Ni concentration reduces the heat of mixing between Mo and the Cu-Ni matrix, while varying the Mo concentration determines whether the lever rule is obeyed during SPD [19]. X-ray diffraction on these samples (Fig. 5 and Table 1) shows the two principal trends that the solubility of Mo in the Cu-Ni-Mo alloys (i) increases with increasing Ni concentration for fixed Mo concentration and (ii) increases slowly with Mo concentration for fixed Ni concentration. These results confirm our expectation that by lowering the heat of mixing between Mo and the Cu-rich matrix, the solubility of Mo can be enhanced. The sizes of Mo precipitates in the $\text{Cu}_{84}\text{Ni}_{08}\text{Mo}_{02}\text{Ag}_{06}$ and $\text{Cu}_{84}\text{Ni}_{07}\text{Mo}_{03}\text{Ag}_{06}$ alloys, 8 ± 3 nm and 10 ± 4 nm, respectively, are somewhat smaller than Mo precipitates in both the corresponding pre-milled Ni-Mo alloys, 11 ± 3 nm and 13 ± 3 nm, and the $\text{Cu}_{89}\text{Mo}_{05}\text{Ag}_{06}$ sample, 12 ± 5 nm, but well within the variances of the distributions. The particle sizes and densities for all compositions are reported in Table 2.

3.2. High pressure torsion (Cu-Mo, Cu-Mo-Ni, Cu-Mo-Ag)

Complementary experiments were performed on Cu-Mo, Cu-Ni-Mo, Cu-Ag-Mo alloys using HPT. Unlike the ball milling experiments, the HPT work was performed on initially solid solution alloys, prepared by magnetron sputtering, as described above. These samples, unlike those from ball milling experiments, contained very little oxygen, less than what could be detected by EDX or XRF, < 0.04 at.%. Past work on similarly prepared films of Cu-Nb showed similarly low oxygen contamination, 0.03 at.% [17]. The HPT experiments, moreover, were performed at significantly lower temperatures, -78°C , using dry ice as the coolant.

TEM images of the as grown $\text{Cu}_{95}\text{Mo}_{05}$ films, Fig. 7, show that the alloy films have columnar grains, with an in-plane grain size of ~ 150 nm. No evidence of Mo precipitation is observed, either in bright field images, or in electron diffraction (Fig. 7). Similarly, no Mo peaks are observed by X-ray diffraction (Fig. 8). Fig. 8 does show, however, a large shift in the Cu (111) peak to lower angles relative to pure Cu (111); in this case, the shift is due to Mo in solution and not a Ag additive (as used for ball milling). With the assumption that Mo is completely dissolved in Cu in the as grown samples, we obtain a molar volume for Mo in FCC Cu equal to $V_{\text{Mo}}^{\text{Cu}} \approx 1.47 \cdot V_{\text{Cu}}^{\text{Cu}}$, where V_B^A is the molar volume of component B in the matrix A [see Supplementary Information Table S1]. We used this value, above, in estimating the solubility of Mo in Cu in the ball milling experiments. For reference, the calculated molar volume of pure FCC Mo is $V_{\text{Mo}}^{\text{Mo(FCC)}} \approx 1.36 \cdot V_{\text{Cu}}^{\text{Cu}}$ [31] thus suggesting a positive volume of mixing, as expected for strongly immiscible alloys.

X-ray diffraction scans shown in Fig. 8(a) and (b) were collected at different locations along the specimen radius to investigate the effect of strain, starting from the center (zero strain). At the center position, no Mo peaks are observed, but the Mo (110) peak does begin to appear with increasing radius, i.e., increasing strain, illustrating Mo precipitation and growth under low-temperature, torsional straining. The small Ag (111) peak at zero strain is due to the ~ 40 nm Ag film coating the sample; it gradually disappears with increasing radial position (i.e., strain), as Ag enters solution. Also observed in Fig. 8 is an increase in texture in the films with increasing strain, as illustrated by the increasing relative intensities of the Cu(111) to Cu(200) peaks. Similar texture development appears common during HPT of polycrystalline materials [32]. The solubilities of Mo in Cu, calculated from the shifts in the X-ray peaks, are plotted in Fig. 9 as a function of strain. After a strain of

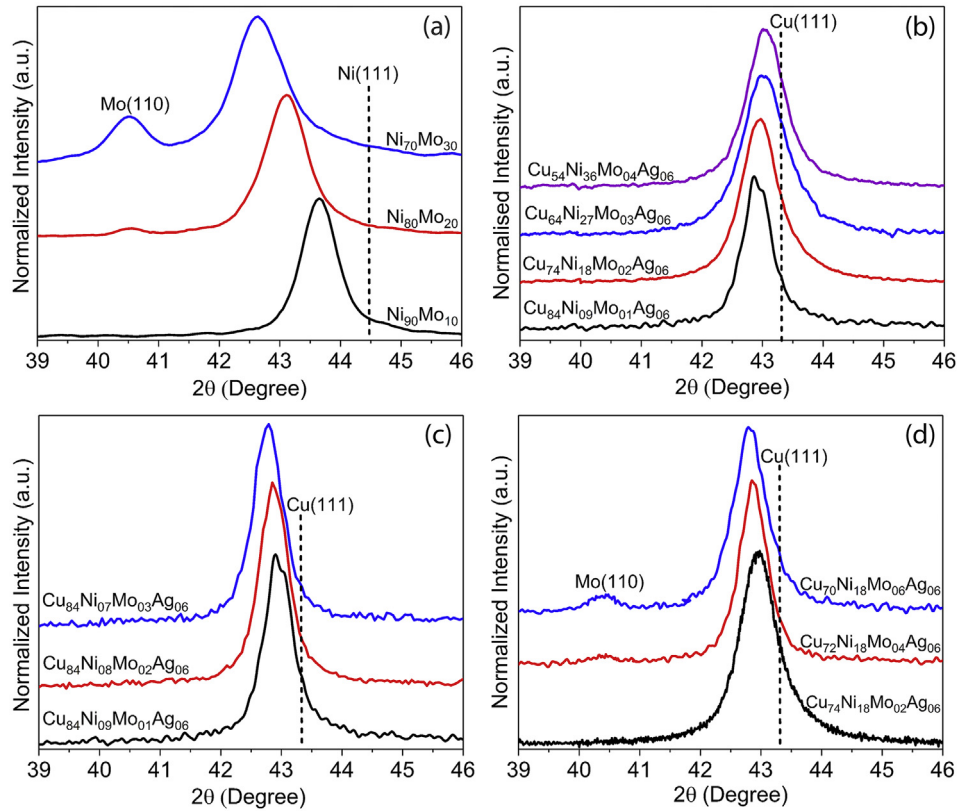


Fig. 5. X-Ray diffraction patterns for: (a) Ni-Mo binary alloys; (b) Ternary Cu-Ni-Mo-Ag with fixed Ni:Mo; (c) with varying Ni:Mo ratio; (d) with fixed Ni concentration. The relative shift of peaks is indicative of Mo solubility in the Ni and Cu-Ni-Ag matrices.

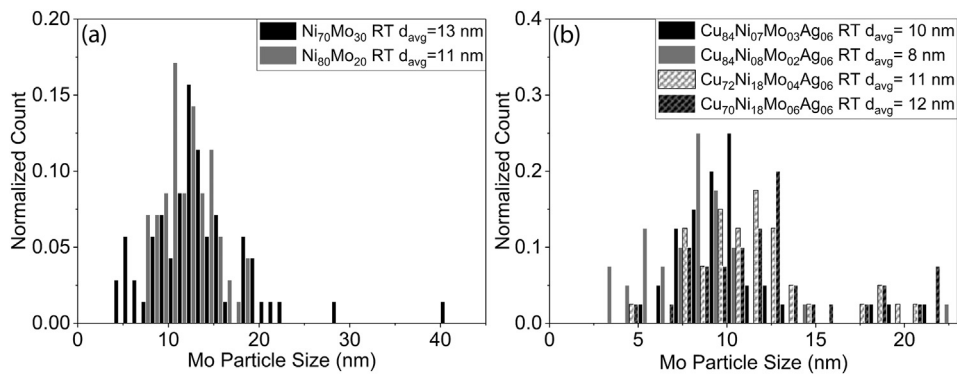


Fig. 6. Mo particle size distribution for binary Ni-Mo alloys with Ni:Mo ratios, 7:3 and 8:2 (a); particle size distributions for ternary Cu-Ni-Mo-Ag alloys with Ni:Mo ratios of 7:3, 8:2, 18:4 and 18:6 (b).

~5,000, the solubility of Mo in the nominally Cu-5 at.% Mo alloy approaches a steady state value, $X_{Mo(sat)}^{Cu} = 0.8\%$, this value appears insensitive to Mo concentration. The addition of 10 at.% Ni to the Cu-Mo alloy significantly increases the solubility, to $X_{Mo(sat)}^{Cu-10\%Ni} = 2.3\text{ at.}\%$. Increasing the Ni concentration to 20 at.% further increases the Mo solubility to $X_{Mo(sat)}^{Cu-20\%Ni} = 2.7\text{ at.}\%$. These findings are similar to those for ball milling, although a slightly higher solubility of Mo in Cu is obtained after HPT (see Table 1). This small difference is close to the uncertainty of the ball-milled results, although it might also be explained by the lower temperature of the HPT processing [see e.g. Ref. [17]], or possibly other factors such as the small amounts of oxygen, Ag and Fe in the ball-milled powders. Fig. 9 also shows that in contrast to the addition of Ni to the Cu-Mo

alloy, the addition of 18 at.% Ag to the Cu-5 at.% Mo alloy has very little effect on the Mo solubility.

TEM characterization of these films after low-temperature HPT shows significant changes in the microstructure. First, the initial columnar grain structure is lost and the grain size is reduced to ~50 nm, as shown in the TEM micrograph in Fig. 7(b). Of greater interest here is the observation of a high density of Mo precipitates, see Fig. 7 (b,c,d) and Table 2. The size distributions of the Mo precipitates in the different alloys are reported in Fig. 10; the average size of the precipitates is $\sim 10 \pm 3$ nm for all of the alloys, in good agreement with the ball-milled samples. For the Cu₉₅Mo₀₅ alloy, the precipitate size is found to increase along the radius, from 3 nm at 1.5 mm ($\epsilon \sim 900$), to 8 nm at 2.5 mm ($\epsilon \sim 2000$) and finally

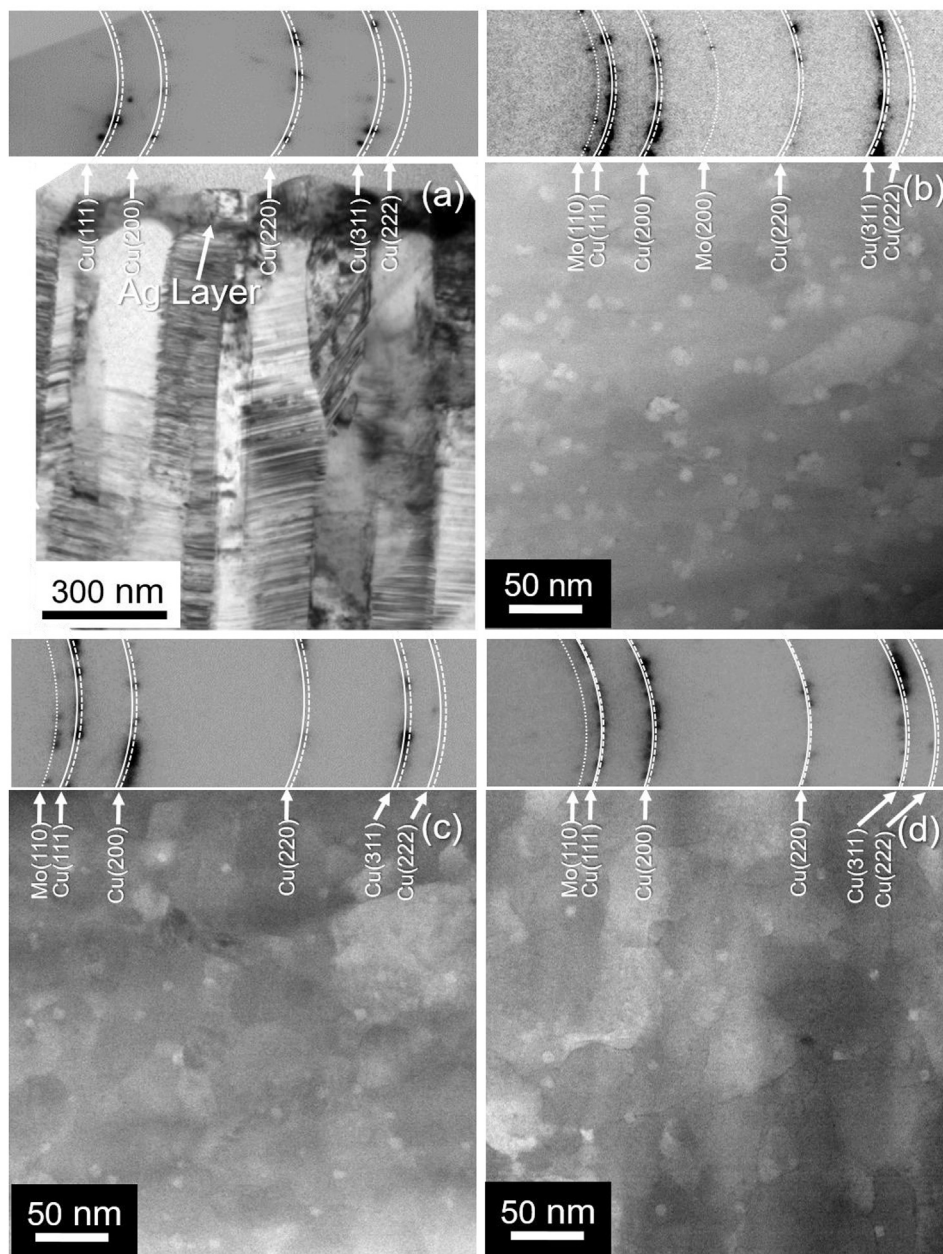


Fig. 7. (a) BF image of the as grown $\text{Cu}_{95}\text{Mo}_{05}$ film with indexed SAD pattern showing: no Mo diffraction rings, indicative of a single phase Cu-Mo solid solution. After HPT processing, phase separation is observed; a clear indication of Mo precipitates is seen in HAADF images: (b) $\text{Cu}_{95}\text{Mo}_{05}$, (c) $\text{Cu}_{85}\text{Ni}_{10}\text{Mo}_{05}$ and (d) $\text{Cu}_{75}\text{Ni}_{20}\text{Mo}_{05}$.

11 nm at 3.5 mm ($\epsilon \sim 6000$) (see Supplementary Information, Fig. S2). The precipitates seem preferentially located at grain boundaries, although with a grain size of 50 nm and an average precipitate size of ~ 10 nm, most precipitates would contact grain boundaries even if randomly distributed. EDX measurements were performed to estimate the Ni concentration in the Mo precipitates. This was done by comparing the ratio of Cu to Ni signals acquired with the electron beam directed on a single Mo precipitate in the matrix with the Cu/Ni ratio in a region not containing precipitates (See Supplementary Information Fig. S3). By assuming that the Mo precipitates contain negligible amounts of Cu, and estimating the relative thicknesses of the sample and precipitate, we conclude that in agreement with the x-ray diffraction results, very little Ni is dissolved in the Mo precipitates, ~ 5 at.%. We emphasize that since Mo atoms precipitate out of the Cu-Ni solution in these HPT

experiments, Ni can be included within the Mo precipitates without necessarily deforming the Mo precipitates. This is unlike ball milling experiments. Thus, the steady state microstructure for the HPT and ball-milled samples are very similar, illustrating that in these dilute Cu alloys, that it is not sensitive to the initial state of the alloy.

Lastly, we examined the structure of the Mo precipitates in the Cu and Cu-Ni matrices using HRTEM. Fig. 11 presents a series of such images. Most of the Mo precipitates are equiaxed, or nearly so, and they show no signs of fracture or amorphization at the Cu/Mo interface. These images also reveal that the Mo precipitates can be a single grain, or they can comprise two or three Mo grains. In some cases, the grains can be highly aligned such as P1 and P2 in Fig. 11(d); notice that no misfit dislocations are observed at the grain boundary. In this case, the boundary plane is (100) with (110)

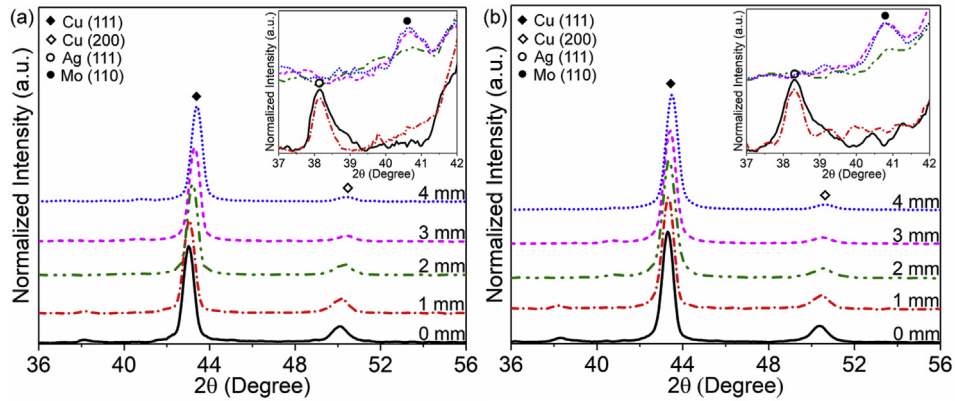


Fig. 8. X-Ray diffraction data for (a) $\text{Cu}_{95}\text{Mo}_{05}$ and (b) $\text{Cu}_{75}\text{Mo}_{05}\text{Ni}_{20}$ at different radial positions. The insets show that the Ag peak disappears while Mo peak emerges as strain increases.

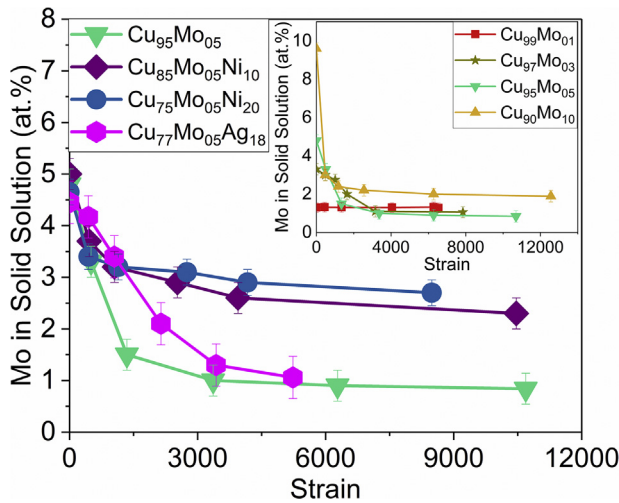


Fig. 9. Mo solubility versus strain for $\text{Cu}_{95}\text{Mo}_{05}$, $\text{Cu}_{85}\text{Mo}_{05}\text{Ni}_{10}$, $\text{Cu}_{75}\text{Mo}_{05}\text{Ni}_{20}$ and $\text{Cu}_{77}\text{Mo}_{05}\text{Ag}_{18}$. Shown in the inset is the solubility versus strain for various binary Cu-Mo alloys.

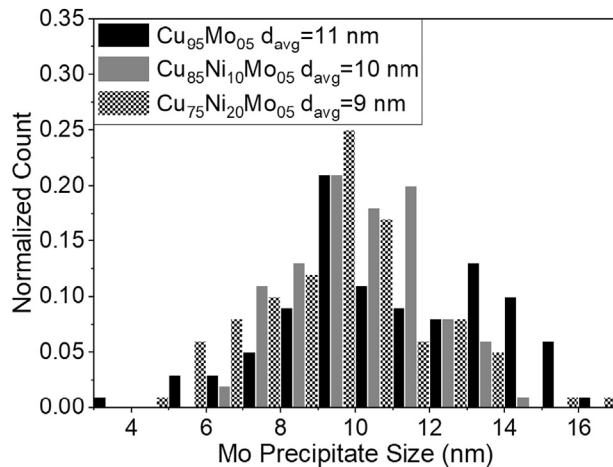


Fig. 10. Mo precipitate size distribution for $\text{Cu}_{95}\text{Mo}_{05}$, $\text{Cu}_{85}\text{Mo}_{05}\text{Ni}_{10}$ and $\text{Cu}_{75}\text{Mo}_{05}\text{Ni}_{20}$.

planes of both precipitates are perpendicular to each other, possibly forming a small angle twist boundary. Another example of low angle grain boundary can be seen in the Supplementary Information, Fig. S4, where boundary misfit dislocations are highlighted. These images suggest that the two precipitates can rotate into a low energy configuration as they come into contact. It is noteworthy that similar relaxation behavior was observed in MD simulations of two particle sintering [33]. Precipitate P3 in Fig. 11(d), on the other hand, has not yet come into contact with either P1 or P2, and it shows no special orientation w.r.t P1 and P2.

3.3. Molecular dynamics simulation

The main results of our simulations are shown in Figs. 12–14. Fig. 12 shows the change of Ω for several Cu(1)–Cu(2) alloys with different heats of mixing as a function of strain during cycling shearing. Notice for the alloy with $\alpha = 1.11$ eV/atom that the final state of order, Ω_{SS}^{SPD} , is independent of the initial state, i.e., random or phase separated. In Fig. 13(a and b) the steady state SRO parameter, Ω_{SS} , in the sheared alloy is plotted (bold line) as a function of the heat of mixing parameter, α . Here, α is the heat of mixing (or heat of solution) per mole at infinite dilution of Mo in Cu, (i.e. Cu(3) in Cu(1)). Following Ref [18] we also plot the equilibrium values of Ω as a function of temperature. As noted above, values of Ω_{eq} were obtained using a hybrid Monte Carlo/Molecular Dynamics code (MC/MD), as implemented in LAMMPS [34]. For both $\text{Cu}(1)_{90}\text{Cu}(3)_{10}$ and $\text{Cu}(1)_{75}\text{Cu}(3)_{25}$ deformed at 100 K, the SRO corresponds well to samples equilibrated at an effective temperature of ~ 6500 K. Only the 25 at.% Cu(3) samples with a heat of mixing = 1.6 eV/atom and both samples with a heat of mixing = 2.0 eV/atom deviate from this trend, tending toward lower effective temperatures. It is noteworthy that it is only in these samples that Cu(3) precipitates are observed in the sheared specimens. Lastly, we plot in Fig. 14 the equilibrium solubility of Mo (i.e., Cu(3)) in Cu and Cu-Ni alloys at various temperatures. For this plot, we selected interaction potentials, $\text{Cu}(i) - \text{Cu}(i \neq j)$ to reproduce the heats of solution for Mo in Cu, Ni in Cu and Mo in Ni.

4. Discussion

The present experiments on Cu-Mo alloys are in broad agreement with a number of past investigations showing that highly immiscible alloys do not homogenize during low temperature SPD. Various explanations for this behavior have been suggested, for a recent review see Ref. [11], however, these explanations tend to be highly specific to microscopic deformation processes rather than to

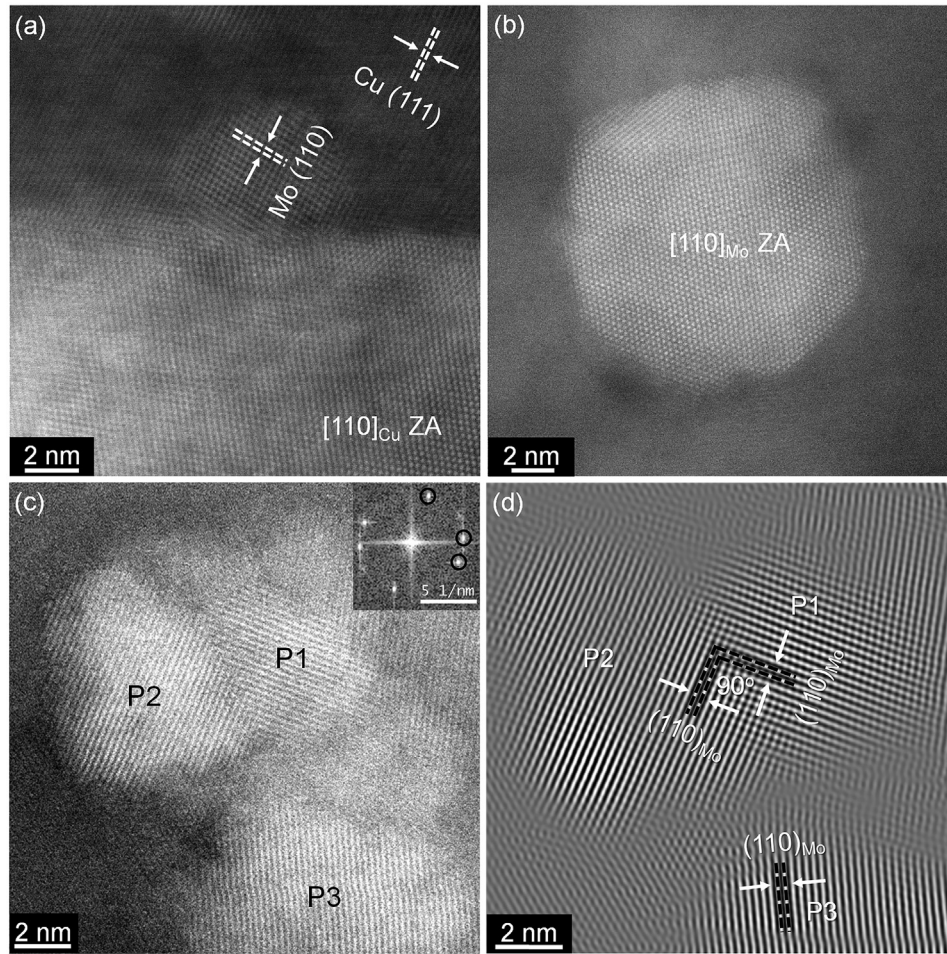


Fig. 11. HRSTEM images of Mo precipitates in Cu₉₅Mo₀₅: (a) precipitate residing at a Cu grain boundary shows alignment with the matrix but no evidence of an amorphous phase at the precipitate/matrix interface. Mo precipitate can be nearly spherical with a single orientation (b), as well as a combination of different orientations (c, d). Precipitate P1 and P2 in (c) and (d) have rotated to a low energy configuration, where diffraction spots from FFT of (c) shown in inset were used to determine the orientation relationship shown in (d).

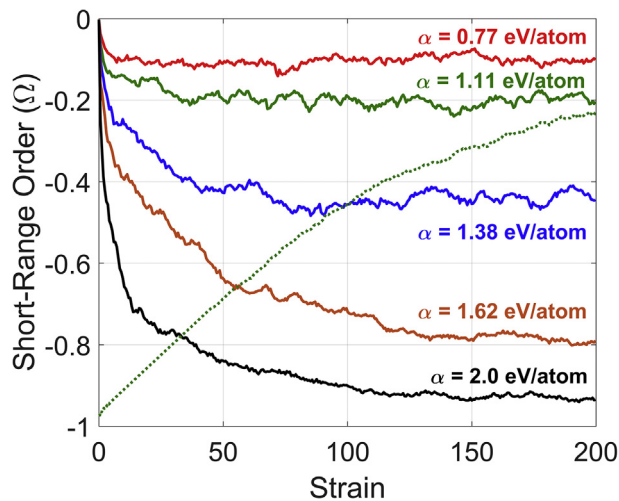


Fig. 12. Short range order parameter vs biaxial compressive strain for Cu(1)-Cu(2) alloys with different heats of mixing.

a more generalized understanding or theory. One of the findings of the current study, that these alloys approach the same steady state microstructure independent of their initial state, suggests that our

results may fit into a more general scheme. In this vein, we now discuss whether the effective temperature model of Martin, as modified by Lund and Schuh [18] and ourselves, provides a quantitative description of the SPD process in our model Cu-Mo alloy systems. We point out from the outset of this discussion that an effective temperature model is unlikely to explain all of our results since we observe that the Mo precipitates approach a unique steady state size, R_{SS} . This was also observed for Cu-Nb [16,17]. Since the current effective temperature model includes no information on length scales, R_{SS} should be infinite.

We begin our evaluation by estimating the effective temperature from the current experiments using the solubility data for binary Cu-Mo alloy. Here we use the HPT data, since it is more accurate as well as being performed at lower temperatures. Since the solubilities of Mo in Cu during HPT (or ball milling) are ~ 1 at. %, we use a dilute solution approximation to estimate the effective temperature, i.e.,

$$X_{Mo}^{Cu} = \exp \left[- \left(\Delta G_{Mo}^j + \alpha \right) / RT_{eff} \right] \quad (5)$$

where ΔG_{Mo}^j is the molar free energy of pure Mo in phase “j”, i.e., FCC or liquid, α is the partial molar heat of mixing of Mo in Cu at infinite dilution, and X_{Mo}^{Cu} is the measured solubility. The value of the heat of solution of Mo in FCC Cu, unfortunately, has not been

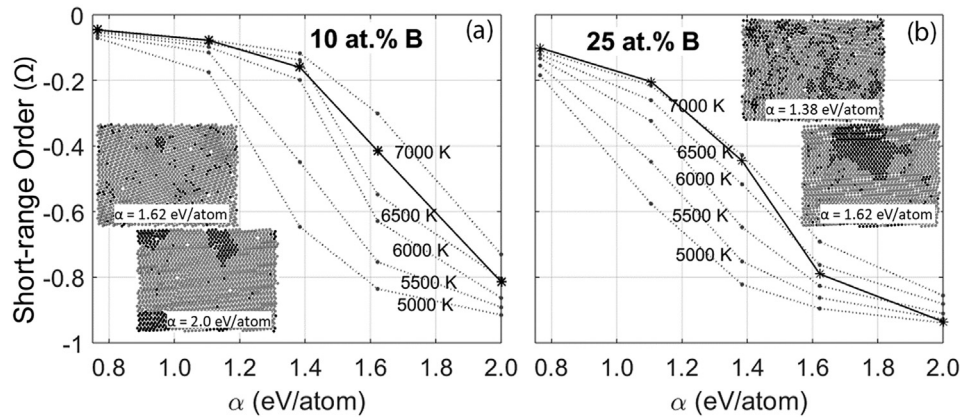


Fig. 13. Steady-state values of SRO from MD (solid line) and MC (dotted lines) at the indicated temperatures as a function of heat of mixing parameter for 10% (a) and 25% (b) nominal solute composition. Insets show the steady-state microstructure obtained from the MD shearing simulations at the indicated heat of mixing value.

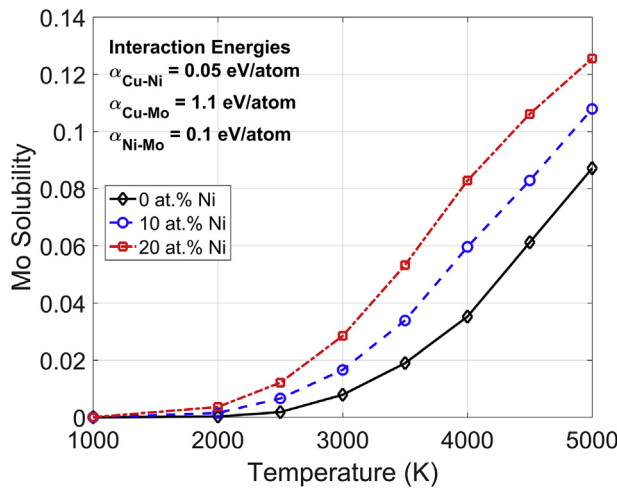


Fig. 14. The equilibrium solubilities of Mo as a function of temperature for ternary Cu-Ni-Mo alloys with different Ni concentrations. Interactions energies from: $\alpha_{\text{Cu-Ni}}$ Ref. [35]; $\alpha_{\text{Ni-Mo}}$ Ref. [36]; $\alpha_{\text{Cu-Mo}}$ (see below).

measured, and so we first estimate it here using a calculated phase diagram. For the liquid Cu-Mo solution, we use the data from Ref. [37] (viz, 1.9% solubility at 2173 K and 2.5 at.% solubility at 2373 K), approximate the molar free energy of Mo in the melt as, $\Delta G_{\text{Mo}}^m = \Delta H_{\text{Mo}}^m(T_m - T)/T_m$, and obtain $\alpha = 65$ kJ/mol for the liquid. We use this value of α for the FCC phase in eq. (5), along with the value $\Delta G_{\text{Mo}}^{\text{FCC}} = \Delta H_{\text{Mo}}^{\text{FCC}}(T = 0) = 41$ kJ/mol obtained from DFT calculations [31] and our experimental value from the HPT experiments, $X_{\text{Mo}}^{\text{CuFCC}} = 0.008$. This yields $T_{\text{eff}} = 2,650$ K. We repeated this same procedure for Cu-Nb, for which the thermodynamic data is somewhat more reliable [38]. From Table 3, $\Delta G_{\text{Nb}}^{\text{CuFCC}} + \alpha = 82$ kJ/mol, and from low temperature HPT experiments ref [17] $X_{\text{Nb}}^{\text{CuFCC}} = 0.025 \pm 0.005$. This yield $T_{\text{eff}} = 2700 \pm 100$ K. We note that using the interatomic potential for Cu-Nb derived in Ref. [38], we obtain $\Delta G_{\text{Nb}}^{\text{CuFCC}} + \alpha = 96$ kJ/mol; this value yields $T_{\text{eff}} = 3100$ K.

Table 3
Calculated solubilities for different Cu binary alloys. ΔG_X^{FCC} are from Ref. [31].

	Mo	Nb	V	Ta	Cr	W	Co	Ag	Ni
α (kJ/mol)	65 [37]	63 [38]	117 [45]	[46]	42 [47]	100 [48]	30 [47]	15 [49]	10 [50]
$\alpha + \Delta G_X^{\text{FCC}}$ (kJ/mol)	106	96	141	164	80	146	30	15	10
X_{sat}	0.014	0.021	0.0034	0.0014	0.04	0.003	SS	SS	SS

Within our level of approximation, these values of T_{eff} for Cu-Nb, are in good agreement with that obtained for Cu-Mo.

We next consider the HPT results for the Cu-Ni-Mo system. In the introduction we suggested that $(\text{Cu}_{90}\text{Ni}_{10})_{95}\text{Mo}_{05}$ could be considered a “virtual” binary alloy, similar to $\text{Cu}_{95}\text{Mo}_{05}$, but with an approximately 10% smaller heat of mixing, owing to the replacement of Cu-Mo bonds with Ni-Mo bonds [see e.g. 55]. While this estimate is reasonable for dilute alloys, it does neglect short range order. We thus used the MC/MD simulations, plotted in Fig. (13), with the data in Table 3, to calculate the Mo solubility in the ternary alloys at 2700 K. We obtain $\Delta G_{\text{Mo}}^{\text{CuFCC}} + \alpha = 96$ kJ/mol and find $X_{\text{Mo}}^{\text{CuFCC}}(10\% \text{ Ni}) = 0.016$. For the 20% Ni alloy, we find similarly, $X_{\text{Mo}}^{\text{CuFCC}}(20\%) = 0.024$. The enhancement for the 20% Ni alloy is in very good agreement with the HPT experiment (Fig. 9); the agreement for the 10% Ni alloy is somewhat less satisfactory, but also reasonable. The Mo solubility in the HPT $(\text{Cu}_{82}\text{Ag}_{18})_{95}\text{Mo}_{05}$ alloy, on the other hand, shows little change from that for the Cu-Mo binary alloy, as expected, since Ag-Mo bond energies are not significantly different from Cu-Mo bond energies. While we have not explicitly evaluated the ball milling data, the overall behavior is consistent. These various results, therefore, suggest that the modified effective temperature model described by eq. (2(b)) and (3) provides a consistent explanation of the enhanced solubilities of strongly immiscible alloys.

We next examine if the result $T_{\text{eff}} \sim 3000$ K for ball milling or HPT of Cu alloys is consistent with the behavior of other immiscible Cu alloys. We begin by estimating the solubilities of these binary systems using the thermodynamic data reported in Table 3 and assuming a regular solution model. Although there is considerable uncertainty in some of these data, particularly for the more immiscible alloys, and the regular solution model may underestimate solubilities, the trends should be correct. Notably, Co and elements with lower heats of mixing are known to form complete solid solutions with Cu at $T_{\text{eff}} \sim 3000$ K: Cu-Ag [25], Cu-Ni [39], and Cu-Co [40], while Cr, and elements with higher heats of mixing, show very little solubility at this temperature, < 2 at.-%: Cu-Mo (this work), Cu-Nb [17,41], Cu-Cr [42], Cu-Ta [43,44]. It should be

Table 4

Nanoindentation hardness of selected alloys. The last column 4mm/SS represents the hardness of the processed alloys in steady state (SS).

Sample	Hardness (GPa)				
	Center	1 mm	2 mm	3 mm	4mm/SS
Cu ₉₅ Mo ₀₅	3.97 ± 0.22	4.34 ± 0.2	4.58 ± 0.25	4.69 ± 0.12	4.71 ± 0.35
Cu ₈₅ Ni ₁₀ Mo ₀₅	4.45 ± 0.12	4.56 ± 0.23	4.82 ± 0.31	4.99 ± 0.25	5.13 ± 0.21
Cu ₇₅ Ni ₂₀ Mo ₀₅	4.88 ± 0.18	5.05 ± 0.28	5.3 ± 0.24	5.26 ± 0.53	5.68 ± 0.31
Cu ₇₇ Ag ₁₈ Mo ₀₅	4.74 ± 0.16	5.23 ± 0.22	5.13 ± 0.24	5.76 ± 0.23	5.83 ± 0.19
Cu ₈₉ Mo ₀₅ Ag ₀₆ (powder)					4.81 ± 0.31
Cu ₇₂ Ni ₁₈ Mo ₀₄ Ag ₀₆ (powder)					5.62 ± 0.28

similarly possible to estimate the effect of various ternary additions to the solubilities of these binary Cu alloys, although we are unaware of any such experimental data, aside from the data presented here.

One of the assumptions in the above analysis is that the effective temperatures of the different alloys are the same. This seems reasonable since the alloys are mostly dilute. Previous work suggested, however, that the relative hardness of the alloy components in elemental form could affect mixing during SPD [11,51]. We, therefore, checked the nanohardness of the various Cu-Ni-Mo and Cu-Ag-Mo alloys. The results are reported in Table 4, primarily for the HPT alloys, since the effective temperatures were deduced from these samples, but also for the ball-milled alloys in steady state (SS). While the additions of Ni and Ag in both increase the hardness by ~20%, the two alloying components have similar effects on hardness, but very different effects on the Mo solubility. This suggests that adding Ni to the Cu-Mo increases the Mo solubility primarily by reducing the effective heat of mixing of Mo in Cu and not from increasing the hardness of the alloy, i.e., the effective temperature does not appear much affected by small additions of Ni or Ag to Cu. We do not imply that large differences in hardness can never be important, e.g., see Ref. [51]. Ball milling of Al-W, for example, does not result in mechanical alloying despite the large negative heat of mixing in this alloy, showing that in this (extreme) case (and milling not performed at low temperatures) the initial state is important [52].

We next discuss briefly the significance of why the effective temperature deduced from the experiments deviates rather significantly from that determined by the MD simulation. Recall the simulations deduce an effective temperature $T_{eff} = 6500$ K. The most obvious explanation for this difference, of course, is the enormous difference in the strain rates, $\sim 10^9$ s⁻¹ for the MD compared to ~ 10 s⁻¹ for the HPT. Ashkenazy et al. [19] indeed reported, for example, that a factor of 2 increase in the MD strain rate increased the effective temperature in amorphous Cu-Ta alloys by ~ 200 °C (at ~ 2000 °C). Possibly increasing the strain rate results in higher stresses [53] thus enabling atoms to overcome larger barriers, but extrapolating this result an additional seven or eight orders of magnitude is not possible. Perhaps more relevant to the present work is our inability to simulate Cu-Mo directly, since Mo precipitates are BCC and they have a high shear modulus. One consequence of the BCC structure, as was demonstrated previously by MD simulations of shearing Nb and V precipitates in Cu [28], is that dislocations do not transfer across the precipitate/matrix interface, at least not for precipitates with radius greater than ~ 2 nm. This implies that the ballistic diffusion coefficient, D_b , which transports Mo atoms across the Cu/Mo interface and into the matrix, is reduced. On the other hand, these same MD simulations revealed that the rate of atomic rearrangements at the Cu/Nb and Cu/V interfaces, which are incoherent, is increased relative to that in the matrix, thereby enhancing relaxation processes. In other words, the competition between ballistic events driving random mixing and relaxation events leading to demixing is weighted more

heavily in these systems toward relaxation. Within the effective temperature model, this lowers T_{eff} . In the current formulation of T_{eff} , this competition is included in the “process temperature”, T_0 . For our MD simulations using Cu(1) and Cu(3), dislocations can glide both in the matrix and in the precipitates, thereby increasing D_b and decreasing relaxation events. This seemingly also explains why precipitates do not form during shearing of Cu(1)-Cu(3) alloys, until the heat of mixing is raised to values far higher than the heat of solution in Cu-Mo, since T_{eff} is higher. This discussion, of course, raises questions about the use of T_{eff} in inhomogeneous systems, since as seen in the Appendix, T_{eff} is defined locally. Such a discussion, however, extends beyond the scope of the present work.

The ball milling results on Ni-Mo are not a major focus of this work, but the low solubility of Ni in the Mo precipitates warrants brief discussion, as it shows the mixing is not symmetric in the two terminal phases. We can explain this result, at least qualitatively within the modified effective temperature model, by using the thermodynamic data from Ref. [36]. The main point is that the partial molar free energy of Mo in dilute Ni alloys is rather small, 10 kJ/mol for dilute FCC Ni-Mo alloys, but it is 33 kJ/mol for dilute BCC Mo-Ni alloys. With an effective temperature of 3000 K, for example, the solubility of Ni in Mo, $X_{Mo}^{Ni} \sim 0.10$ while there is virtually complete solubility of Mo in Ni, in good qualitative agreement with the experiments. This result is thus consistent with the effective temperature model.

Lastly, we discuss the mechanism of precipitate growth in the initially Cu-Mo and Cu-Ni-Mo solid solutions. Previous atomistic simulations revealed that forced atomic mixing in crystalline materials occurs predominantly by dislocation motion [24]. An important consequence of this dynamics is that the rate of relative displacements of atoms and clusters will be nearly the same, independent of cluster size, at least while the dislocation glide distance is much larger than the cluster size. This dynamics is very different from that for thermally activated diffusion where the diffusion of clusters scales inversely with the fourth power of the cluster radius, or even slower [54]. Precipitate growth, therefore, will be dominated by cluster agglomeration. This mechanism of growth explains very well the HRTEM images in Fig. 11, showing that many Mo precipitates contain grain boundaries, with each grain of comparable size. Further discussion of the agglomeration process will be presented elsewhere.

5. Conclusions

A highly immiscible, dilute Cu-Mo alloy, characterized by a high positive heat of solution was subjected to SPD using ball milling and high-pressure torsion. Phase evolution was assessed by monitoring the Mo solubility as a function of strain. Ternary alloying elements, Ni and Ag, were added to systematically change the effective heat of solution of Mo in the Cu-rich matrix; Ni lowering the heat of solution and Ag leaving it unchanged. A detailed microstructure investigation in conjunction with MD/MC simulations leads to the following main conclusions:

1. The steady state solubility of Mo in Cu during SPD is ~1 at.%. At higher concentrations of Mo, the excess Mo forms equiaxed nanoprecipitates, ~10 nm in diameter, with the density of nanoprecipitates increasing with average Mo concentration.
2. The addition of Ni to the Cu-Mo alloy leads to an increased solubility of Mo in Cu, whereas the addition of Ag had no measurable effect. While additions of Ni and Ag also effect the hardness of the alloys, the changes were very similar for the two alloying additions, suggesting that the heat of solution is the dominant controlling parameter.
3. The results are consistent with the effective temperature model of Lund and Schuh [18] (which was re-derived here). For Cu-Mo and Cu-Nb the effective temperature during low-temperature SPD is ~3000 K. This value of T_{eff} is consistent with the experiments on several other immiscible Cu alloys.
4. MD simulations demonstrated that the effective temperature model also reproduces the short range order in immiscible Cu alloys undergoing SPD, but with T_{eff} ~6500 K. While the much

$$\begin{aligned}
 J &= -M(T) \cdot \nabla \mu(T) - M_b(T_0) \cdot \nabla \mu(T_0) \\
 J &= -c\bar{c} \left\{ \left[\frac{D(T)}{kT} \nabla \mu(T) \right] + \left[\frac{D_b(T_0)}{kT_0} \nabla \mu(T_0) \right] \right\} \\
 &= -c\bar{c} \left\{ \nabla \left[\frac{D(T)}{kT} \mu(T) + \frac{D_b(T_0)}{kT_0} \mu(T_0) \right] \right\} \\
 &= -c\bar{c} \left\{ \frac{\partial^2}{\partial c^2} \left[\frac{D(T)}{kT} G(T) + \frac{D_b(T_0)}{kT_0} G(T_0) \right] \nabla c \right\}
 \end{aligned} \tag{A1}$$

with $\bar{c} = 1 - c$

In the above equations, M , D , μ and G correspond respectively to atomic mobility, diffusivity, the chemical potential driving inter-diffusion, and Gibbs mixing free energy, the subscript “b” referring to ballistic processes. The last two equalities are obtained by assuming that D 's are independent of the local concentration. The Gibbs free energies are then expressed using a regular solution model with an interaction coefficient Ω :

$$\begin{aligned}
 \frac{D(T)}{kT} G(T) + \frac{D_b(T_0)}{kT_0} G(T_0) &= \frac{D(T)}{kT} [\Omega c\bar{c} + kT(c \log c + \bar{c} \log \bar{c})] + \frac{D_b(T_0)}{kT_0} [\Omega c\bar{c} + kT_0(c \log c + \bar{c} \log \bar{c})] \\
 &= \left[\frac{D(T)}{kT} + \frac{D_b(T_0)}{kT_0} \right] \left\{ \Omega c\bar{c} + [D(T) + D_b(T_0)] \left[\frac{D(T)}{kT} + \frac{D_b(T_0)}{kT_0} \right]^{-1} (c \log c + \bar{c} \log \bar{c}) \right\}
 \end{aligned}$$

higher strain rate employed in MD relative to experiments might explain this high temperature, it was suggested that heterogeneities, viz. Hard, incoherent nanoprecipitates, might also explain this result.

5. HRTEM revealed that Mo nanoprecipitates in Cu are equiaxed during SPD, with no evidence of particle deformation or fracture. This morphology suggests that the steady state size of the nanoprecipitates is controlled by an erosion process. HRTEM also showed that the Mo nanoprecipitates could be single grain or multigrained. Several grain boundaries were observed to be low energy boundaries, suggesting that the precipitates can rotate into low energy configurations during the agglomeration process.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actamat.2019.03.023>.

Appendix

In this appendix we briefly present a derivation of a modified effective temperature, based on Martin's approach [5] but assuming that the ballistic mixing takes place at a high but finite temperature T_0 . The interdiffusion flux is assumed to be the sum of two terms, a thermally activated dynamics at temperature T and a forced dynamics at temperature T_0 .

One recognizes in the curly brackets of the last equation the expression for the Gibbs free energy of mixing of the alloy, but evaluated at an effective temperature defined by:

$$kT_{eff} = [D(T) + D_b(T_0)] \left[\frac{D(T)}{kT} + \frac{D_b(T_0)}{kT_0} \right]^{-1}$$

We then define the forcing intensity as:

$$\gamma_b^{T_0} = \frac{D_b(T = T_0)}{D_{th}(T)} = \frac{D_b^{T_0}}{D_{th}}$$

leading to a more compact expression of this effective temperature:

$$kT_{eff} = \left[1 + \gamma_b^{T_0} \right] \left[\frac{1}{kT} + \frac{\gamma_b^{T_0}}{kT_0} \right]^{-1} = kT \left[1 + \gamma_b^{T_0} \right] \left[1 + \frac{T}{T_0} \gamma_b^{T_0} \right]^{-1} \tag{A2}$$

One sees that when $T_0 \rightarrow \infty$ this expression reduces to Martin's effective temperature.

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